



Leandro Di Bella

SOFTWARE ENGINEER – (3D) COMPUTER VISION ENGINEER – MACHINE LEARNING ENGINEER

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“Deployable autonomous perception with temporal consistency and robustness under distribution shift.”

Summary

PhD candidate looking for a full-time opportunity. Specialized in computer vision and AI for autonomous driving and autonomous navigation for drones, with expertise in detection, multi-object tracking, and vision-language models (VLMs). Currently working on multi-agent motion forecasting using flow-based generative AI for autonomous perception and modular sensor fusion for autonomous drones. Seeking to contribute to Europe’s research and innovation ecosystem by advancing AI solutions for intelligent systems.

Experience

VoxelSensors (research project) COMPUTER VISION RESEARCHER	Jan. 2026 – Present Brussels, Belgium	Sensor fusion and localization for autonomous drone systems.
MACQ Mobility (research project) COMPUTER VISION RESEARCHER	Jan. 2023 – Jan. 2026 Brussels, Belgium	Detection, 3D multi-object tracking, scene understanding, and motion forecasting for autonomous vehicles using generative AI.
Mappx FOUNDER	Aug. 2025 – Present Brussels, Belgium	Built a map-centric social travel app (Flutter) with FastAPI backend and PostgreSQL.
MACQ Mobility COMPUTER VISION INTERN	Aug. 2022 – Oct. 2022 Brussels, Belgium	Developed and integrated instance segmentation on NVIDIA Jetson TX2 (Python/C++).
Vrije Universiteit Brussel TEACHING ASSISTANT	~2022 – 2025 Brussels, Belgium	Teaching assistant for Machine Learning and Big Data Processing (2.5 years).

Education

ETRO, Vrije Universiteit Brussel (VUB) PhD in Engineering Sciences (AI & Computer Vision)	Brussels, Belgium 2023 – Present
Solvay Brussels School Advanced Master in Industrial & Technological Management	Brussels, Belgium 2025 – 2027 (ongoing)
Bruface (ULB/VUB) M.Sc. Electrical Engineering (Information Technology Systems)	Brussels, Belgium 2020 – 2023

Selected publications

HybridTrack: A Hybrid Approach for Robust Multi-Object Tracking L. Di Bella, Y. Lyu, B. Cornelis, A. Munteanu IEEE RA-L / ICRA (2025)	ReferGPT: Towards Zero-Shot Referring Multi-Object Tracking T. Chamiti, L. Di Bella, A. Munteanu, N. Deligiannis CVPR Workshops (2025)
FlowS: One-Step Motion Prediction via Local Transport Conditioning L. Di Bella, A. Munteanu, B. Cornelis arXiv preprint (2026)	LAM3D: Leveraging Attention for Monocular 3D Object Detection D.-A. Sas, L. Di Bella, Y. Lyu, F. Oniga, A. Munteanu IEEE MMSP (2024)

Skills

Programming	Python, C++, CUDA, PyTorch, TensorRT, ONNX, FastAPI, Flutter/Dart
ML / Perception	Detection, instance segmentation, monocular 3D, 2D/3D tracking (MOT), Kalman filtering, temporal consistency
Vision-language & GenAI	Grounding / referring tracking, multimodal reasoning, generative motion forecasting
Systems / Deployment	Jetson, Docker, GitHub CI/CD, Azure cloud services, Firebase, PostgreSQL
Product	Mapbox, backend ops, shipping & iteration
Academic service	Reviewer: TIP, RA-L, ICRA, ICIP, IET